Assignment: Image Generation (VAE, DDPM, and LDM)

Tasks:

1. Develop a Variational Autoencoder (VAE) to generate synthesized images of your favorite object that are visually similar to real images captured using your phone.
2. Generate five synthesized images by passing five noise vectors/matrices sampled from a normal distribution with a mean of 5 and a given variance through the decoder of your trained VAE.
3. Compare the effects of Binary Cross-Entropy (BCE) loss and Mean Squared Error (MSE) loss when used as the reconstruction loss during VAE training. Discuss their effects on training performance and the quality of the generated images.
4. Express the KL-divergence loss formula in terms of the mean and variance of the latent Gaussian distribution.
5. Develop and train a Denoising Diffusion Probabilistic Model (DDPM) to generate synthesized images of the same object. Explain the forward diffusion process, reverse denoising process, network architecture, training procedure, and image generation process.
6. Generate five synthesized images using the trained DDPM by starting from five noise matrices sampled from a standard normal distribution and performing the reverse diffusion process.
7. Develop and train a Latent Diffusion Model (LDM) to generate synthesized images of the same object. Explain the role of the VAE encoder and decoder, latent-space diffusion, U-Net, timestep embedding, and conditioning mechanism used in the LDM.
8. Generate five synthesized images using the trained LDM by starting from five noise matrices sampled in the latent space and performing the reverse diffusion process, followed by decoding the generated latent representations into images using the VAE decoder.
9. Compare the synthesized images generated by the VAE, DDPM, and LDM in terms of:
    - Image quality
    - Sharpness
    - Diversity
    - Computational cost
    - Training complexity
    - Image-generation time
10. Discuss the advantages and limitations of VAE, DDPM, and LDM and explain why diffusion in latent space can be more computationally efficient than diffusion directly in image/pixel space.